

One sequence model for every HSP pull-down: what ESMC-6B predicts, and what it does not

9 October 2026

Summary

The task. One multi-task MLP on ESMC-6B protein embeddings (the 6-billion-parameter ESM3-family model) and its annotated sparse autoencoder (SAE). It predicts every non-drug quantity measured in the HSP pull-downs:

Kind	Targets	Runs
Baseline pull-down, no perturbation	pull-down level	GA_20, GA_22, GA_24, FS73, FS76 (HSPB1); GA_33 (DNAJA1, DNAJB11)
Abundance	lysate level, supernatant level	GA_22, GA_24, GA_33
Abundance-free binding	bound:free partition; enrichment over input	GA_22, GA_24 (partition); GA_33 (enrichment)
Effects	salt	GA_20, GA_22, GA_24
	temperature	GA_33
	Mg ²⁺	FS76

That makes 33 targets over 11,541 proteins. Drug effects are excluded.

Every number is out-of-fold. Folds are grouped by 30% sequence identity, so no homolog of a test protein is ever trained on. Each number is read against its own split-half ceiling.

Question	Answer
How much does sequence predict?	Mean R^2 / ceiling 0.27-0.28 over all 33 targets. Abundance 0.51; abundance-free binding 0.33; baseline pull-down 0.26; temperature 0.24; salt 0.23; Mg 0.17
Is it better than a linear read-out?	Yes, slightly: +0.028 [0.023, 0.034] over ridge on the best layer; most on salt (+0.039)
Does reading residue by residue help?	No. A learned attention-pooling front end over 7.4 M residue states equals the mean-pooled model: +0.000 [-0.003, +0.004]
Does one shared model help?	Marginally. Multi-task beats 33 single-task models on 22 of 32 targets, but by 0.003 R^2 on average
Do conditions share one representation?	Yes. A trunk trained without a run predicts that run nearly as well as one trained with it. The exceptions are the condition types no other run contains: heat (0.26 $\rightarrow$ 0.19) and Mg (0.15 $\rightarrow$ 0.10)
What predicts the pull-down with no perturbation?	Mostly how much protein there is. Measured abundance plus T_m beats sequence for every baseline. Sequence plus those measurements is best: R^2 0.30-0.54
Fold changes from sequence alone, no input information?	R^2 0.12-0.27 per perturbation (rho 0.35-0.52), calibrated; top-10 % responders recovered 2.5-4x above chance; same accuracy as the full model (-0.002 [-0.007, +0.002])
What does sequence add beyond abundance?	Salt and Mg responses: +0.056 and +0.045 R^2 / ceiling over measured abundance and T_m alone

Question	Answer
Regressing out the baseline	Salt and Mg responses are almost untouched (83-99% of their variance remains). Heat is not: the baseline accounts for about half of the 43 °C response and half of what sequence predicts about it
What the SAE features say	Pull-down and abundance: structured enzymes up, phosphorylated disordered Ser/Thr-rich regions down. Abundance-free binding, salt and Mg: membrane-helix features. Heat: structured α/β enzyme cores

The bottom line. Sequence predicts about a quarter of what these measurements reliably contain. The limit is the information in the representation: an MLP over 94 architectures, attention pooling and multi-task sharing all hit the same plateau. The no-perturbation pull-down is mostly abundance and stability. The salt and Mg responses are where sequence carries information that the measured lysate properties do not.

1. Data, targets and features

Proteins. The union of every run is 11,570 proteins, of which 11,541 have a sequence and all features. Embeddings and SAE features already existed for 9,879. I computed the rest in one pass of ESMC-6B (1,662 proteins, under 1 minute on one RTX card).

A control re-embedding of 60 stored proteins reproduced them: median r 1.0000. One protein moved slightly with batch context (r 0.98). Embedded alone it sits between its two batched versions, so this is bf16 numerics, not a pipeline error.

Targets (cm02). 33 per-protein quantities, each with a split-half reliability across proteins (the ceiling):

Kind	n	Ceiling range
Abundance (GA_33 input, GA_22/24 supernatant)	3	0.98-0.995
Baseline pull-down (no perturbation)	7	0.98-0.998
Abundance-free binding (GA_22/24 partition, GA_33 enrichment)	4	0.99
Salt (GA_20 75/150 mM, GA_22/24 60/120 mM in pull-down, supernatant and partition)	14	0.76-0.99
Temperature (GA_33, 37 and 43 vs 35 °C, both baits)	4	0.64-0.98
Mg ²⁺ (FS76)	1	0.98

Drug arms were excluded throughout, with one exception: FS76's Mg effect is averaged over the lithium doses, because lithium has no proteome-wide effect (reliability 0.14). GA_22 and GA_24 effects use only their drug-free arms and carry the drift correction of the GA_22/24 report.

Inputs.

- ESMC-6B hidden states 0, 10, ..., 80, mean-pooled.
- The layer-60 SAE: 16,384 features, max over residues.
- Per-residue final-layer states for the attention model: 7.36 M residues, 37 GB.
- Sequence covariates: length, charge, pI, hydropathy, composition.
- Optionally, measured per-protein properties: lysate abundance here, PaxDb abundance, Meltome Tm.

2. The model and how it was trained

One MLP:

- an input branch per layer (a learned projection) and for the SAE;
- a shared trunk ending in a 64-dimensional latent;
- one linear head per target, reading that shared latent.

The loss is masked MSE over all observed targets, each standardised on the training fold.

Efficient training. Each architecture is trained as a population of 72-144 members, differing in learning rate, weight decay, dropout and seed. All members are one batched computation: every weight carries a member axis, and a vectorised AdamW applies per-member hyperparameters.

All inputs sit on the GPU. Each member is early-stopped on an inner validation fold, and the outer test fold is predicted by the mean of the eight best members (chosen on validation only).

GPU choice. RTX PRO 6000 (96 GB) for everything:

- data plus a population is 2-50 GB;
- the attention model holds 37 GB of residue states plus 15-25 GB of work;
- nothing needed an H100 or H200.

Job type	SM active	Occupancy	Tensor active	DRAM active	Peak memory
MLP populations	0.84-0.95	0.49-0.58	0.04-0.05	0.76-0.87	3-50 GB
Attention MLP	0.95	0.48	0.21	0.66	54-66 GB
ESMC-6B embedding					18.5 GB at 12-17 k residues/s

The populations are memory-bandwidth-bound: the matrices per member are small. Packing more members per step was the lever, and it kept SM activity near the maximum.

Scale. About 10 GPU-hours on at most 8 RTX cards, in roughly three hours of wall clock:

- 94 MLP architectures;
- 7 attention configurations;
- 33 single-task models;
- 6 leave-one-run-out trunks;
- 6 residualisation and covariate models.

I stopped four sweep jobs early. Their SAE-input groups cost 10-20 minutes each and had already shown that the SAE adds nothing to the dense layers.

3. How much sequence predicts

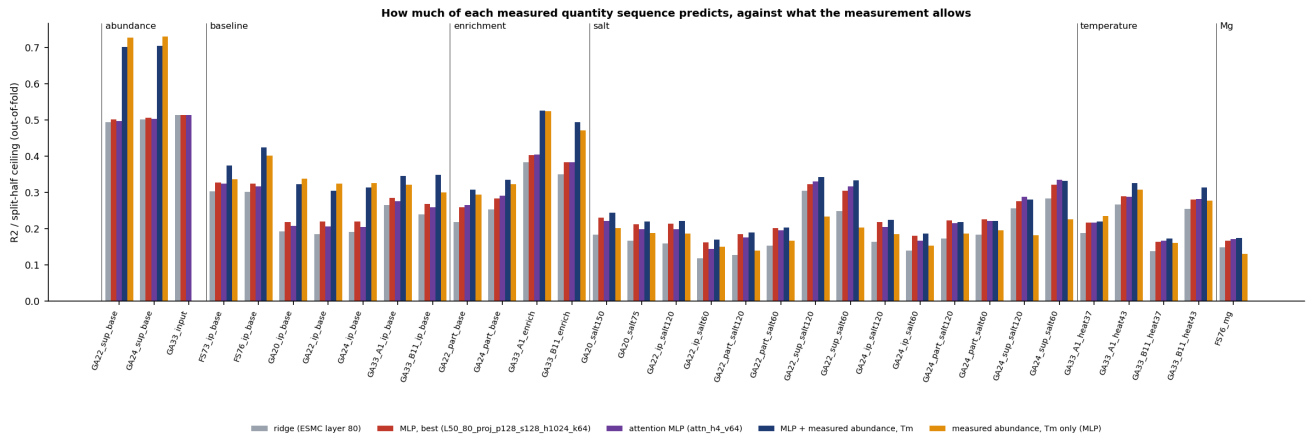


Figure 1: Out-of-fold R^2 / ceiling for every target: ridge on ESMC layer 80; the best MLP; the attention-pooling MLP; the MLP with measured abundance and Tm added; measured abundance and Tm alone.

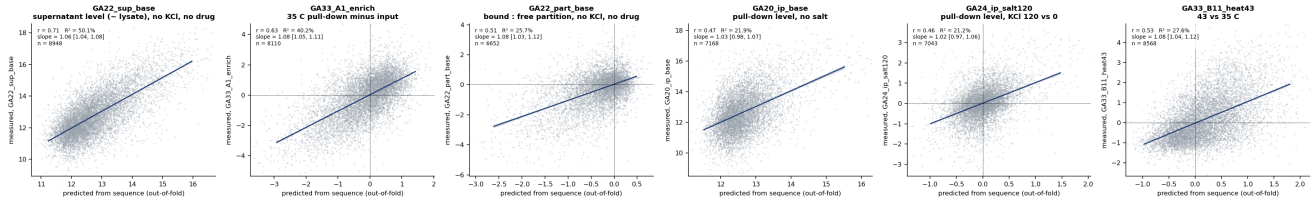


Figure 2: Predicted (out-of-fold) against measured for one target of each kind, with the least-squares line and 95% band.

Mean R^2 / ceiling, all targets, with paired cluster-bootstrap intervals (1,000 resamples of sequence clusters):

Model	All	Abund.	Base.	Enrich.	Salt	Temp.	Mg
Ridge, ESMC layer 80	0.245	0.504	0.240	0.302	0.191	0.213	0.149
MLP, layers 50+80	0.273	0.506	0.261	0.333	0.229	0.238	0.168
Attention MLP	0.273	0.505	0.257	0.336	0.230	0.239	0.172
Abundance + Tm only (32 t.)	0.286	0.729	0.336	0.403	0.186	0.246	0.131
Sequence + abundance + Tm (32 t.)	0.316	0.703	0.348	0.416	0.243	0.258	0.176
Sequence only (32 t.)	0.263	0.497	0.257	0.329	0.228	0.233	0.167

The models marked “32 t.” are trained on 32 targets; the MLP row is the multi-task model with 144 members; the lysate abundance target is dropped because it is the abundance input.

Comparison	Difference in R^2 / ceiling, 95% interval
MLP – ridge	all +0.028 [0.023, 0.034]; salt +0.039 [0.028, 0.050]; enrichment +0.031; abundance +0.002 (n.s.)
Attention – mean-pooled MLP	all +0.000 [−0.003, +0.004]; no kind differs
Sequence + measurements – measurements alone	salt +0.056 [0.042, 0.072]; Mg +0.045 [0.024, 0.065]; baseline +0.012 (n.s.); abundance −0.026
Measurements – sequence	abundance +0.233; baseline +0.079; enrichment +0.074; salt −0.042; Mg −0.036

A plateau. All 94 MLP architectures lie within 0.274-0.276 of the best. That holds across:

- layer sets (50 + 80 or all nine);
- projection widths of 32-256;
- latents of 16-256;
- trunks of 256-2,048;
- 150 or 300 epochs.

The SAE as an input is worse than the dense layers: 0.18 alone, nothing added on top. Attention pooling over residues, at 4, 8 or 16 heads, equals mean pooling. The ensemble’s members are nearly interchangeable (best member vs top-8 mean: $r = 0.9996$).

The limit is what ESMC-6B’s representation carries about these measurements, not the read-out.

The one weakness. Adding sequence to the measured covariates lowers abundance (0.729 $\rightarrow$ 0.703) and a few individual baselines. The combined model’s regularisation does not fully adapt when one input block is far more informative than the other. A residual design (sequence predicting what the covariates leave) would avoid this.

3b. Fold changes alone: each perturbation from sequence, with no input information

This model knows nothing about the input / lysate distribution. One MLP is trained only on the 11 pull-down fold changes, with no abundance, supernatant, partition or baseline targets and no measured covariates: ESMC-6B layers 50 + 80 and sequence covariates in, 11 fold changes out (cm11, run s5).

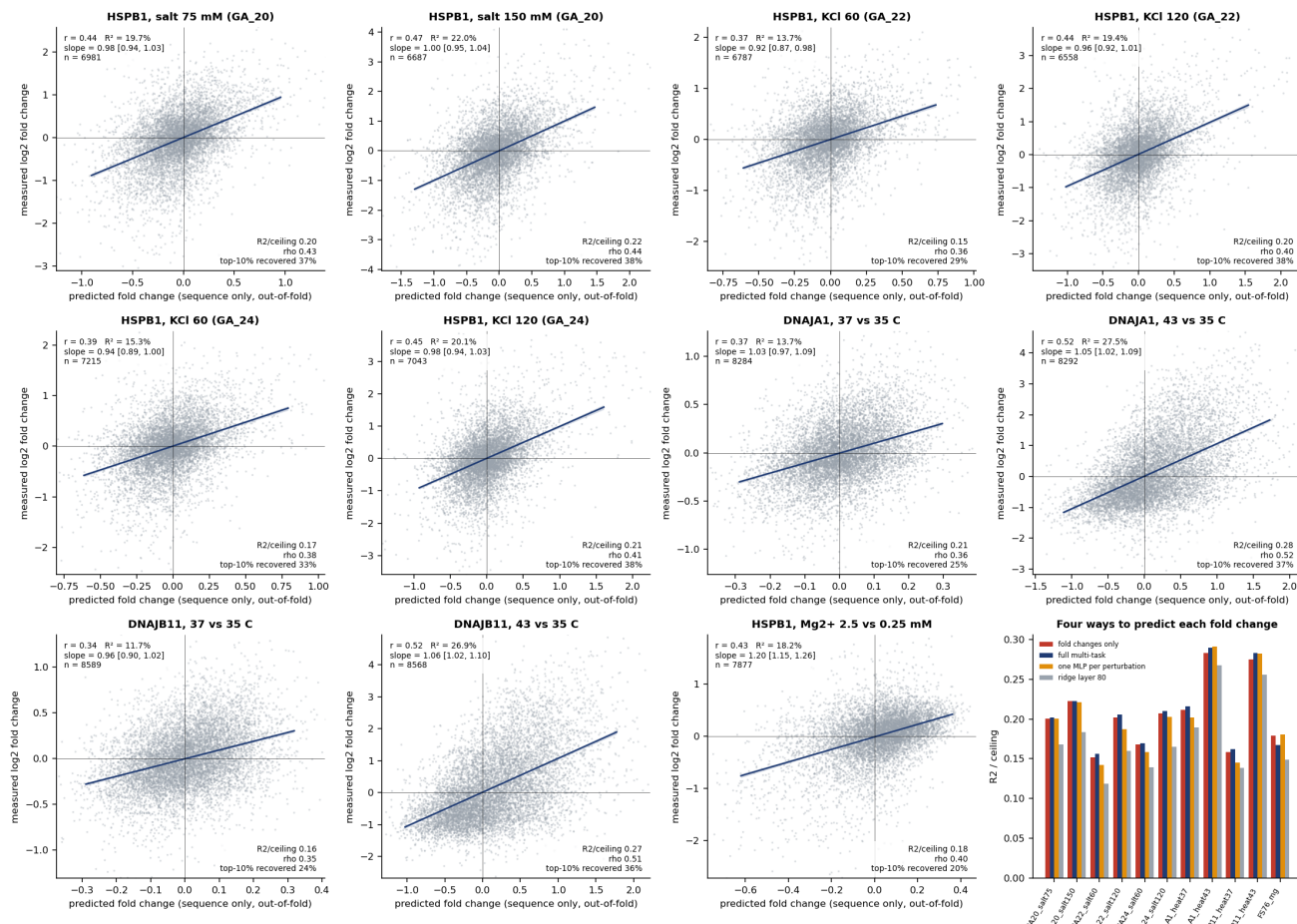


Figure 3: Predicted against measured log2 fold change for every perturbation (sequence only, out-of-fold), with the least-squares line; last panel: the same targets from four models.

Perturbation	Ceiling	R2	R2/ceil.	rho	Top 10 %	Bottom 10 %	Sign
HSPB1, salt 75 mM (GA_20)	0.98	0.197	0.20	0.43	37 %	30 %	79 %
HSPB1, salt 150 mM (GA_20)	0.99	0.220	0.22	0.44	38 %	29 %	80 %
HSPB1, KCl 60 (GA_22)	0.90	0.136	0.15	0.36	29 %	27 %	76 %
HSPB1, KCl 120 (GA_22)	0.96	0.193	0.20	0.40	38 %	27 %	78 %
HSPB1, KCl 60 (GA_24)	0.91	0.152	0.17	0.38	33 %	27 %	76 %
HSPB1, KCl 120 (GA_24)	0.97	0.201	0.21	0.41	38 %	25 %	79 %
DNAJA1, 37 vs 35 C	0.64	0.136	0.21	0.36	25 %	28 %	79 %
DNAJA1, 43 vs 35 C	0.97	0.274	0.28	0.52	37 %	33 %	89 %
DNAJB11, 37 vs 35 C	0.74	0.117	0.16	0.35	24 %	25 %	77 %
DNAJB11, 43 vs 35 C	0.98	0.268	0.27	0.51	36 %	27 %	89 %
HSPB1, Mg2+ 2.5 vs 0.25 mM	0.98	0.176	0.18	0.40	20 %	37 %	77 %

“Top / Bottom 10 %”: of the proteins measured in the top (or bottom) 10 %, the share the prediction also places in its top (bottom) 10 %; chance is 10 %. “Sign”: among the 10 % of proteins with the largest measured change, the share whose predicted change is on the same side of the median.

- **Sequence alone predicts every fold change, modestly:** R2 0.12-0.27 (0.15-0.28 of ceiling), rank correlation 0.35-0.52. The predictions are calibrated (out-of-fold slopes 0.92-1.06; Mg 1.20), so the size of a predicted change means what it says.
- **As a screen it is 2.5-4 times better than chance:** a quarter to over a third of the strongest responders in each direction land in the predicted top or bottom 10 %, and the direction is right for 76-89 % of the strongest. Heat at 43 C is the most predictable perturbation; KCl 60 mM and heat 37 C, the smallest effects, the least. Mg is asymmetric: proteins that fall with Mg are found far better (37 %) than those that rise (20 %).
- **Leaving out the input information costs nothing:** the fold-change-only model equals the full multi-task model (difference in mean R2 / ceiling -0.002 [-0.007, +0.002]), matches one MLP per perturbation (+0.004 [-0.000, +0.008]) and beats ridge (+0.030 [0.022, 0.037]). The abundance and baseline targets did not help the network learn the fold changes, consistent with section 5.3: salt and Mg responses are largely independent of the baseline.

4. What predicts the pull-down with no perturbation

A pull-down level mixes how much protein is in the tube (abundance) with how strongly it binds. Both parts were tested with named, readable properties, alone and with the sequence model (cm07).

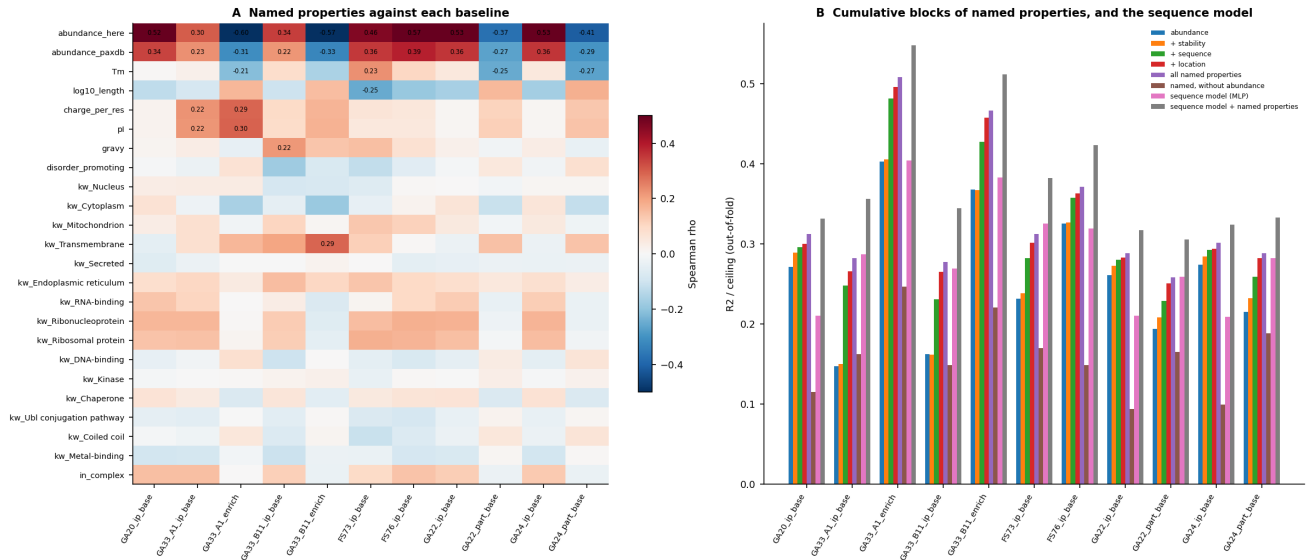


Figure 4: A: Spearman rho of each named property with each baseline. B: out-of-fold R^2 / ceiling as property blocks are added, the sequence MLP alone, and the two combined.

Predictor set	HSPB1 pull-down	DNAJA1/B11 pull-down	Binding over abundance
Abundance (here + PaxDb)	0.23-0.32	0.15-0.16	0.19-0.40
+ Tm, sequence physics, location	0.28-0.36	0.26	0.25-0.49
All named properties	0.29-0.37	0.27-0.28	0.26-0.50
Named properties without abundance	0.09-0.17	0.15-0.16	0.16-0.25
Sequence MLP alone	0.21-0.32	0.27-0.28	0.26-0.40
Sequence MLP + named properties	0.31-0.42	0.34-0.35	0.30-0.54

Values are out-of-fold R^2 . The ceilings here are 0.98-0.998, so R^2 / ceiling differs by under 0.01.

What the properties show:

- **Abundance dominates the raw pull-down.** It correlates with every HSPB1 baseline (rho 0.46-0.57).
- **It reverses for enrichment.** Abundant proteins are less enriched over input (rho -0.57 to -0.60 in GA_33). Capture rises less than proportionally with amount, as found before (slope about 0.35).
- **Abundance-free binding** rises with transmembrane segments and with positive charge (DNAJA1: rho $+0.30$ with pI), and falls with stability (Tm, rho -0.21 to -0.27).
- **Ribonucleoprotein and complex membership** raise the raw HSPB1 pull-down slightly.

Sequence predicts the abundance-free binding about as well as all the named properties together, and the two combined are best. So sequence carries most of what the named properties do, and somewhat more.

5. One representation across conditions

5.1 Leave-one-run-out transfer

Six trunks were each trained without one run's targets. That run's targets were then read out from the frozen latent by ridge, within each fold, since fold models' latents are not aligned with each other.

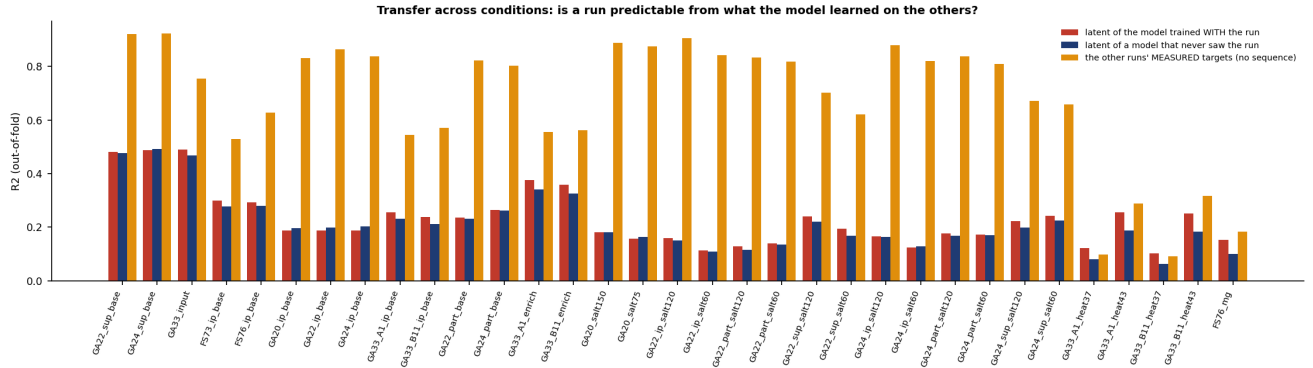


Figure 5: For each target: read-out from the latent of the model trained with the run (red), from a trunk that never saw the run (blue), and, for scale, the prediction from the other runs' MEASURED targets with no sequence (orange).

Run left out	Its targets, read from a trunk that never saw it (with the run)
GA_20 (HSPB1, salt)	baseline 0.20 (0.19); salt 75 / 150: 0.16 / 0.18 (0.16 / 0.18)
GA_22, GA_24 (HSPB1, salt; with supernatant and partition)	within 0.00-0.03 of trained-with for all 18 targets
FS73 (HSPB1)	baseline 0.28 (0.30)
FS76 (HSPB1, Mg)	baseline 0.28 (0.29); Mg 0.10 (0.15)
GA_33 (DNAJA1/B11, heat)	abundance 0.47 (0.49); enrichment 0.32-0.34 (0.36-0.38); heat 43 °C 0.18-0.19 (0.25-0.26)

One shared representation covers every HSPB1 baseline and salt condition: no run teaches it something the others do not. What it loses when a run is left out is a condition type found nowhere else. Heat exists only in GA_33 and Mg only in FS76. The two DNAJ baits also lose a little on their baselines.

The measured cross-run prediction (orange) shows how much the conditions share as data:

- salt responses are 70-93% predictable from the other salt experiments' measurements;
- heat is only 9-34% predictable from anything else measured.

Heat is the condition least like the others.

5.2 Which conditions the model represents along the same axes

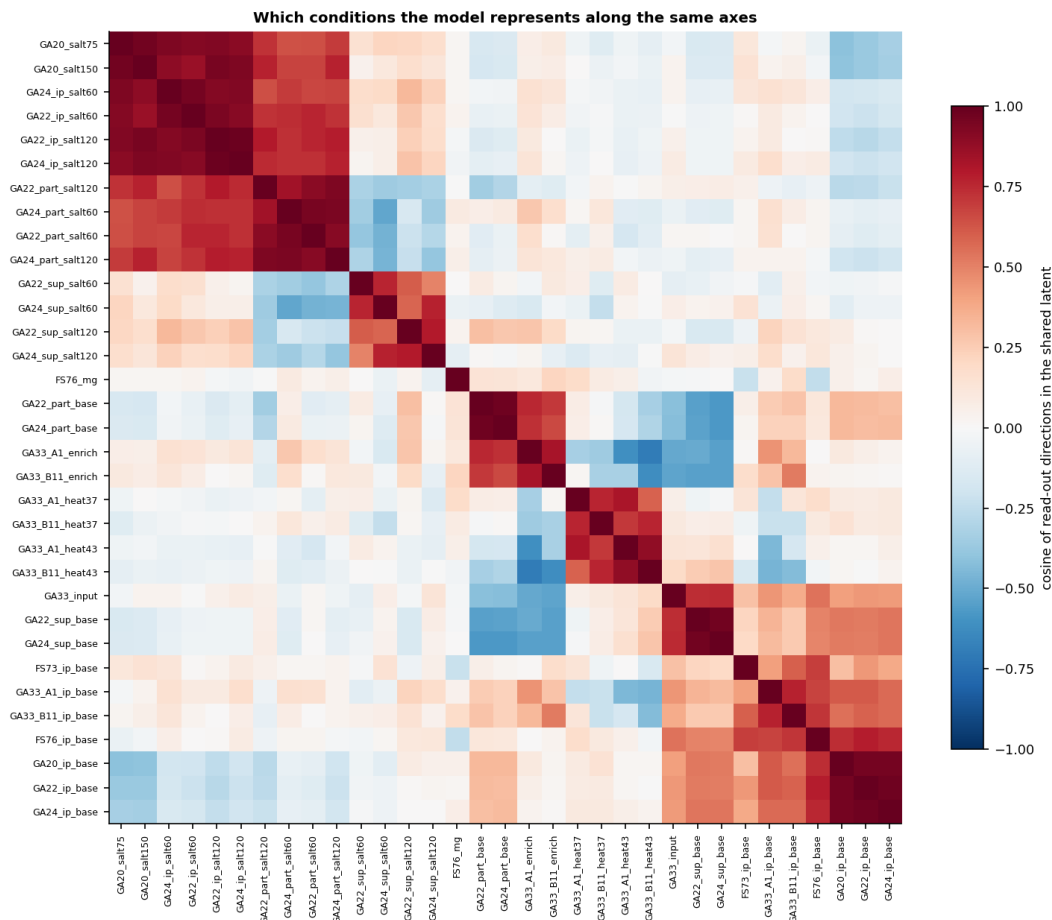


Figure 6: Cosine between targets' read-out directions in the shared latent, clustered.

The latent organises the conditions into a few axes:

- **Salt, pull-down and partition:** one axis shared by all three salt experiments (GA_20, GA_22, GA_24).
- **Salt, supernatant:** a separate axis. Salt changes the free pool along different directions from binding, as the GA_22/24 report found from the data.
- **Abundance and baselines:** an abundance axis carrying the supernatant and input levels, with the baselines close to it.
- **Enrichment and partition:** pointing opposite to abundance.
- **Heat:** an axis of its own.
- **Mg:** aligned with nothing.

5.3 Regressing the baseline out

Two versions, which agree.

(a) In the model. Within each fold, the subspace of the latent that predicts abundance, baselines and enrichment (rank 14) was projected out, and each effect read again. Effects barely change:

- GA_20 salt: 0.16 $\rightarrow$ 0.15;
- heat 43 °C: 0.25 $\rightarrow$ 0.24;
- Mg: 0.15 $\rightarrow$ 0.13.

The effect signal sits in directions separate from the baseline signal.

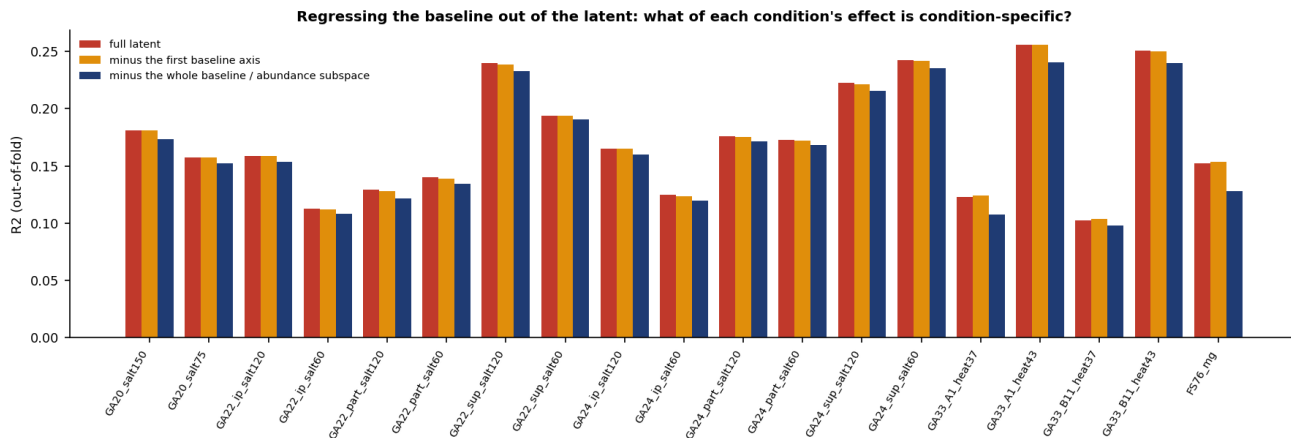


Figure 7: Effect targets read from the full latent, from the latent minus its first baseline axis, and minus the whole baseline / abundance subspace.

(b) In the data. Each effect was residualised on its own run's measured baseline and abundance, fitted on training proteins only, and a fresh MLP was trained on the residuals.

Effect and baseline in one run share their control samples, and therefore their measurement noise. So I repeated it with baselines from an independent experiment: GA_24's for GA_22, FS73's for GA_20 and FS76, the other bait's for GA_33.

Effect	Variance left after removing baseline (own run / independent run)	Sequence R^2 , raw $\rightarrow$ residual (own / independent)
Heat 43 °C, DNAJA1	0.52 / 0.68	0.27 $\rightarrow$ 0.13 / 0.20
Heat 43 °C, DNAJB11	0.52 / 0.52	0.26 $\rightarrow$ 0.13 / 0.10
Heat 37 °C	0.86-0.88 / 0.85-0.97	0.11-0.13 $\rightarrow$ 0.07-0.11
Salt, pull-down (GA_20, GA_22, GA_24)	0.83-0.94 / 0.90-0.94	essentially unchanged
Salt, supernatant	0.83-0.99 / 0.84-0.99	unchanged or higher (0.25 $\rightarrow$ 0.28-0.31)
Mg	0.92 / 0.97	0.17 $\rightarrow$ 0.16 / 0.15

The heat response at 43 °C is about half a starting-level effect. Proteins' 43 °C change depends on their 35 °C level, even when that level comes from another experiment. About half of what sequence predicts about heat goes through that level.

Salt and Mg responses are condition-specific. Almost none of their variance is explained by the baseline, and what sequence predicts about them is untouched by removing it.

6. What the SAE features say

Gradient attribution through an SAE-input model was unstable from fold to fold (r 0.19-0.29). The 16,384 features are highly redundant, and different fold models credit different members of correlated groups. So the features are ranked by a stable statistic instead: Spearman rho with the family-averaged target. Its agreement between two halves of the proteins split by sequence cluster:

Family	Halves r	Top-50 overlap
Abundance	0.93	0.82
Pull-down baseline	0.84	0.78
Bound:free and enrichment	0.90	0.56
Salt	0.73	0.72
Mg	0.79	0.68
Temperature	0.87	0.24

Labels are Biohub’s LLM-generated annotations, which are hypotheses. Each is checked against the UniProt keywords of the proteins on which the feature is active.

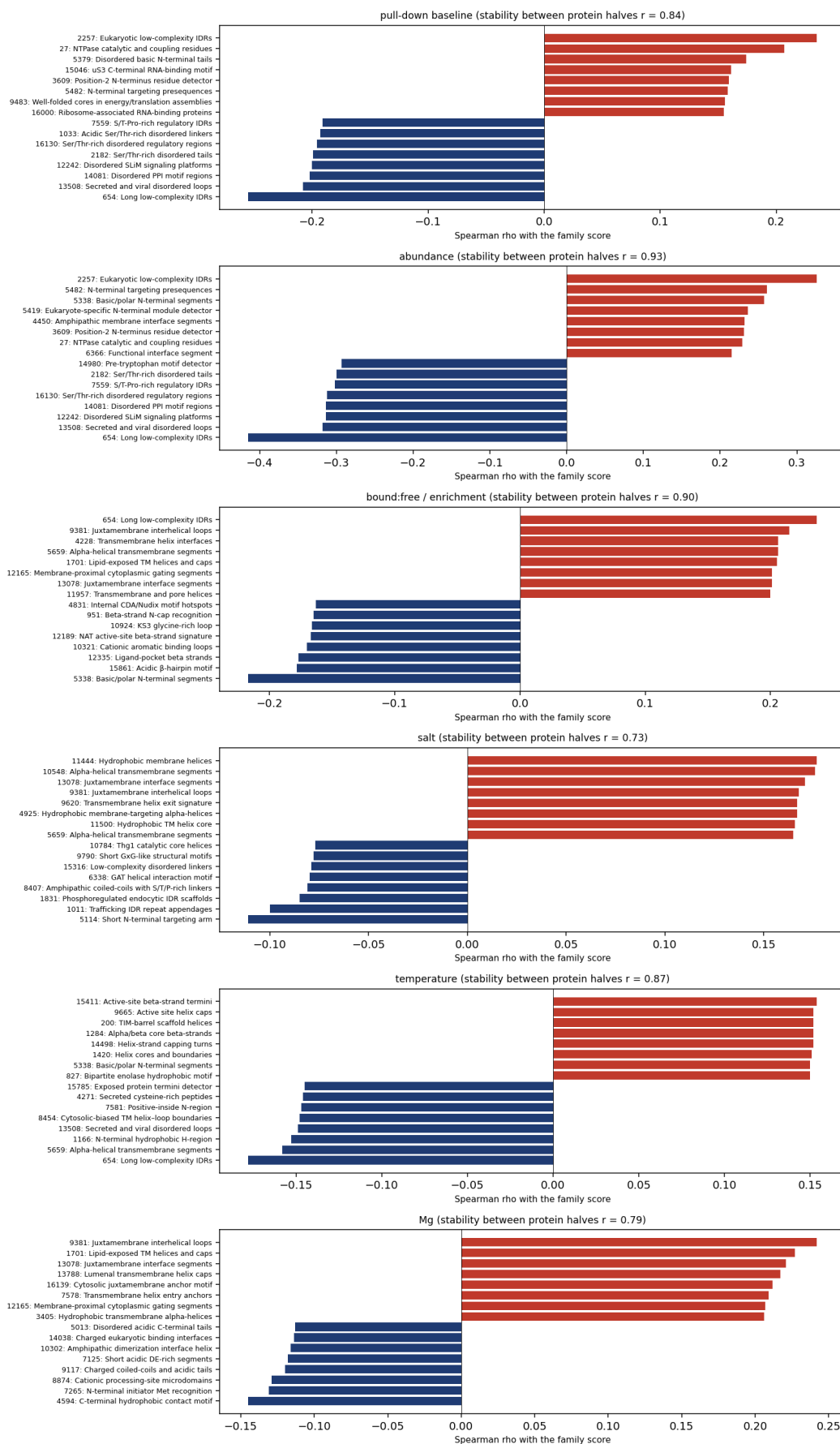


Figure 8: The 8 features most positively and most negatively associated with each family, with their Biohub labels.

Family	Higher with (label; keyword check)	Lower with
Pull-down baseline, abundance	low-complexity IDRs typical of eukaryotic enzymes; NTPase catalytic residues (nucleotide-binding); basic N-terminal tails and ribosomal RNA-binding motifs (ribosomal proteins)	Ser/Thr-rich, S/T-Pro-rich and SLiM disordered regions (59-78% phosphoproteins); secreted disordered loops
Bound:free / enrichment	transmembrane helices, juxtamembrane loops (62-74% membrane proteins); long low-complexity IDRs	enzyme ligand-pocket β -strands and active-site loops (hydrolases); basic N-terminal mitochondrial segments
Salt	hydrophobic and juxtamembrane transmembrane helices (72-82% membrane)	phosphoregulated trafficking / endocytic IDR scaffolds
Temperature (43 °C)	α/β enzyme cores, TIM barrels, active-site helix caps (nucleotide-binding, hydrolases)	membrane, secreted, signal-peptide and disordered features
Mg	transmembrane-helix features (61-75% membrane)	acidic tails, charged coiled coils, nuclear interfaces

Two cautions.

- The membrane axis appears in salt, Mg and enrichment alike. It may describe how membrane proteins behave in a detergent-free lysate (aggregation, insolubility) as much as chaperone binding.
- The heat features (structured α/β enzyme cores) fit thermal unfolding, but the temperature ranking's top features are the least stable of any family (top-50 overlap 0.24). Read them as a class, not as individual features.

7. A proteome-wide prediction table

The five fold models were applied to the 8,836 reviewed human proteins that no run measured; each fold model was trained on 80% of the measured proteins. This adds predictions for every target to the out-of-fold predictions for the 10,777 measured proteins: 19,613 proteins in `data/campaign/proteome_predictions.tsv`.

The expected accuracy is each target's out-of-fold R^2 (0.10-0.49). The table is a prior for picking candidates, not a substitute for measurement.

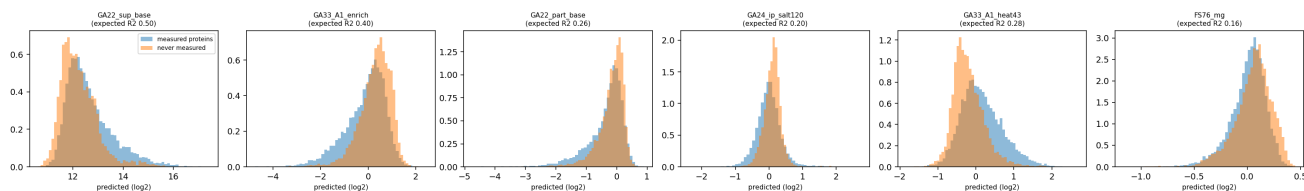


Figure 9: Predicted distributions for measured and never-measured proteins, for six targets.

8. Critical evaluation

1. **One protein-level representation.** ESMC-6B mean-pooled states carry about a quarter of the reliable variance, and nothing tried here moved that. The rest is either not in the sequence (lysate state, complex membership, post-translational state, local concentration) or not in this representation. The measured covariates show the first is substantial for baselines.
2. **Abundance as input or target.** For baselines, measured abundance is the strongest predictor, and sequence partly predicts it. The fair question is what sequence adds beyond abundance. That is small for baselines (+0.012, interval includes 0) and real for salt and Mg.

3. **Shared measurement noise.** In-run residualisation shares control samples between effect and baseline. The independent-baseline version removes that and gives the same answer.
4. **Not every run measures abundance.** HSPB1's FS73 / FS76 and GA_20 have no supernatant or input, so abundance comes from GA_33's lysate and PaxDb. Different lysate preparations add noise.
5. **The SAE is not an accurate input.** It is the most interpretable one, but on its own it predicts less (0.18) than the dense layers. Its feature associations describe the measurements, not the model's predictions.
6. **Bugs found and fixed during the campaign:**
 - a shared row-index file overwritten by later specs, which zeroed the first latent read-out;
 - a drift-covariate aliasing in the linear model on sparse proteins (from the GA_22/24 work);
 - variable shadowing in the per-residue writer, caught by its own completeness assertion.

Each was caught by a check before any number was reported.

9. Files

- **Code:**
 - `cm00_union.py`, `cm01_embed.py`, `cm01b_residues.py` (embeddings, SAE, per-residue states);
 - `cm02_targets.py`, `cm02b_features.py` (targets, features, folds);
 - `cm03_mlp.py` (population MLP, covariates, residualisation), `cm03b_ridge.py`;
 - `cm04_latent.py` (transfer, regress-out, geometry, measured cross-run);
 - `cm05_attrib.py`, `cm05b_assoc.py`, `cm05c_annot.py` (SAE);
 - `cm06_attn.py` (attention pooling), `cm07_baseline.py` (named properties);
 - `cm08_figures.py`, `cm09_stats.py` (bootstrap), `cm10_proteome.py`;
 - `cm_launch.py` and `cm_summary.py` (sweeps);
 - `slurm/cm0*.sbatch`.
- **Results:** `data/campaign/`:
 - `runs/` (every group: config, R^2 , out-of-fold predictions, latents, saved models);
 - `ridge.json`, `cm07_baseline.json`, `cm09_stats.json`;
 - `attrib/` (associations, annotated features);
 - `proteome_predictions.tsv`.
- **Figures:** `reports/figures/campaign/`.